3D-PRINTED ENCLOSURE GUIDE · SKU CQA240229P1

# Raspberry Pi 5 Case & PiEEG Lid

A two-part printable enclosure for a Raspberry Pi 5 with a PiEEG-8 or PiEEG-16 shield: a vented base with port cut-outs and 6 mm extra height for the shield, and a screw-on lid with cable slots for the electrode leads.



| 2 | 96.6 × 65 | 35.5 | 4 × M3 | PLA/PETG |
|---|---|---|---|---|
| Printed parts | mm base footprint | mm base height | Lid screws | Material |

## 01 PARTS AND FILES

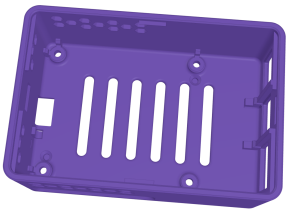
| Part | File | Size (mm) | Volume |
|---|---|---|---|
| **Base** — Raspberry Pi 5 case | RPi_5_case_airvents_slim_ports_+6mm_2.stl | 96.6 × 65.0 × 35.5 | 34.4 cm³ |
| **Lid** — PiEEG-8 / PiEEG-16 | PiEEG_8_16_Lid_V3_screws.stl | 90.2 × 59.6 × 1.8 | 8.5 cm³ |
| CAD source (both parts) | RPi_PiEEG_case_STEP.zip | STEP | — |

STL files are ready to slice; the STEP files can be opened in Fusion, FreeCAD, Onshape or SolidWorks to adapt the design.

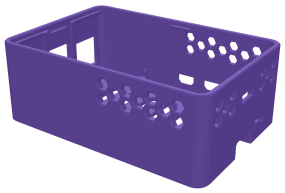
## 02 VIEWS



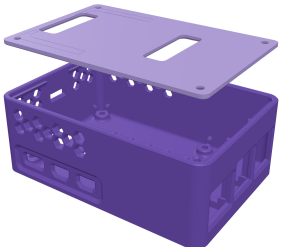
Base — vents and Pi 5 port cut-outs



Base from above — Pi stand-offs, floor vents



Base — rear and side vents



Exploded — lid above the base



Lid — two cable slots, 4 screw holes



Assembled


For research and education only. Not a medical device. Specifications subject to change.

## 03 KEY DIMENSIONS

| | |
|---|---|
| **Base outside** | 96.6 × 65.0 × 35.5 mm, walls with hexagonal vents and floor slots |
| **Shield clearance** | +6 mm height over a standard Pi 5 case, for the stacked PiEEG shield |
| **Lid** | 90.2 × 59.6 × 1.8 mm, recessed into the base rim |
| **Lid screw holes** | 4 × Ø3.26 mm (M3 clearance), 81.2 × 50.6 mm pattern |
| **Cable slots** | 2 × 9.5 × 29.0 mm, 51.3 mm apart — electrode leads and access to the header |

## 04 RECOMMENDED PRINT SETTINGS

| Setting | Base | Lid |
|---|---|---|
| Material | PLA or PETG | PLA or PETG |
| Nozzle / layer height | 0.4 mm / 0.2 mm | 0.4 mm / 0.2 mm |
| Walls / top-bottom | 3 perimeters / 4 layers | 3 perimeters / 4 layers |
| Infill | 15–20 % gyroid | 20 % |
| Orientation | Open side up (floor on the bed) | Flat, top face down |
| Supports | Only under the side port cut-outs if your printer needs them | None |
| Approx. filament | ≈ 30–40 g | ≈ 10 g |

Filament estimates assume PLA (1.24 g/cm³) with the settings above; check your slicer. Use a non-conductive material only — no carbon-filled or metal-filled filaments.

## 05 ASSEMBLY

**1 Print and clean**
Print both parts, remove any supports and check that the port cut-outs line up with your Pi 5.

**2 Fit the Raspberry Pi**
Place the Pi 5 on the four stand-offs in the base, ports facing the cut-outs. Secure with M2.5 screws if required.

**3 Mount the shield**
With the Pi unpowered, press the PiEEG shield onto the 40-pin header until fully seated.

**4 Route the leads**
Pass the electrode leads through the lid slots, then plug them into the electrode header.

**5 Close the lid**
Fit the lid into the rim and fix it with 4 × M3 screws. Do not over-tighten into plastic.

**WARNING The case does not change the safety rules.**
Even in a case, PiEEG-8 must run from a 5 V power bank only while electrodes are on a person, with every connected device on battery.

## 06 RELATED DOCUMENTS

| | | |
|---|---|---|
| **PE8-DS-001**<br>Product Datasheet | **PE8-MD-002**<br>Mechanical & Dimensional Drawing | **PE8-PCB-003**<br>PCB & Manufacturing Specification |
| **PE8-PIN-004**<br>Connector Pinout & Electrode Guide | **PE8-QSG-005**<br>Quick Start & Safety Guide | **PE8-SW-006**<br>Data Interface & Software Guide |